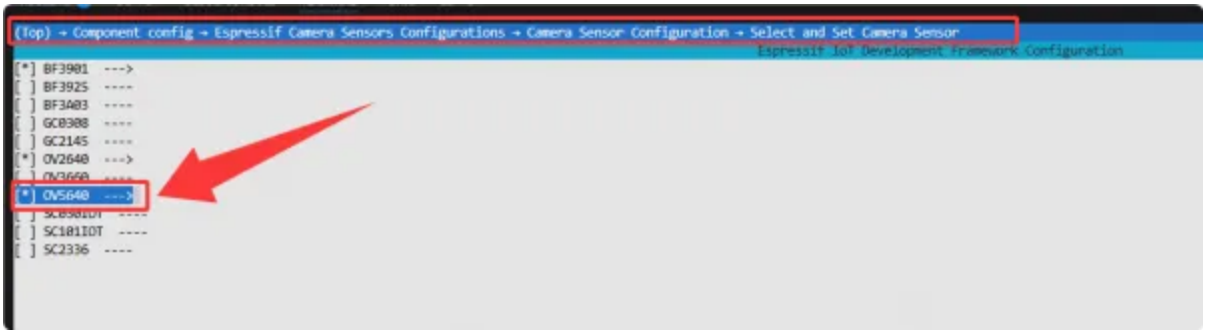
FAQ

Q: What platforms are supported by the AI voice conversation example?

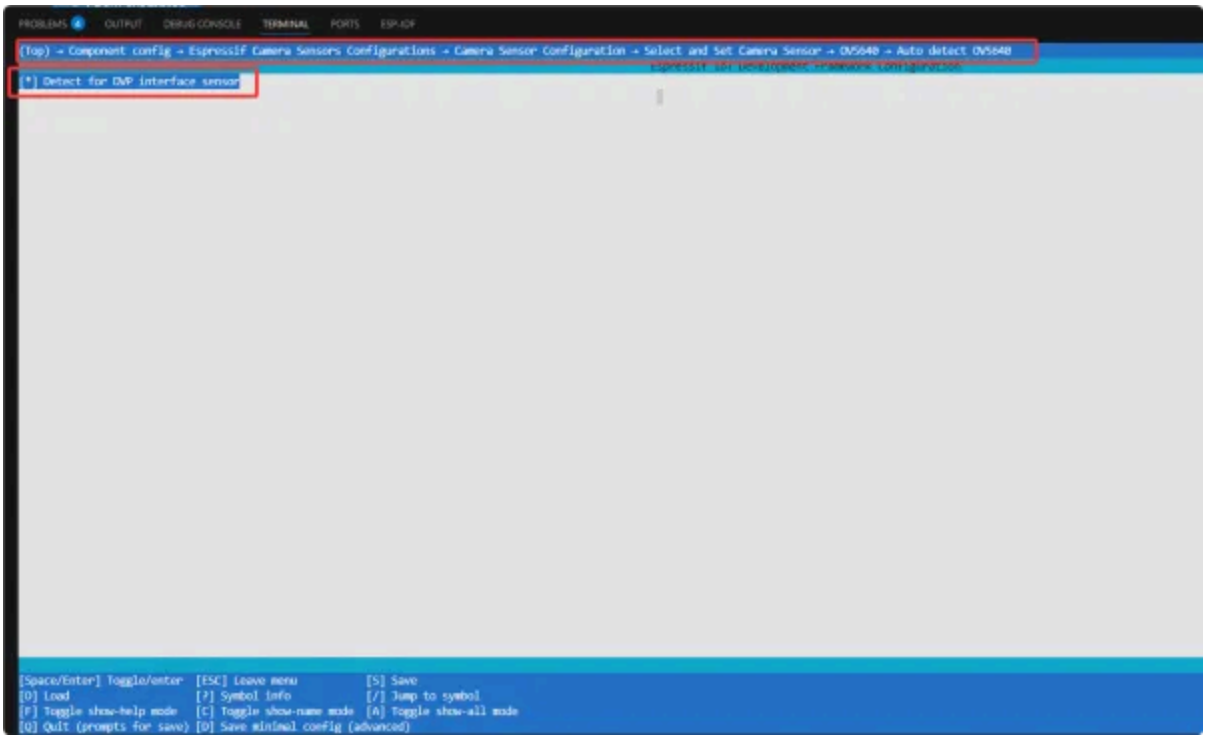
- This product features an onboard ES8311 audio codec chip, onboard speaker, and SMD microphone, enabling voice conversation functionality.
- Integration with various platforms has been verified, supporting mainstream platforms such as Doubao, Wenxin Yiyan, ChatGPT, etc. The examples will be gradually released to ESP32-AIChats

Q: The ESP32-S3-Touch-LCD-3.5B fails to open the camera after flashing Xiaozhi source code?

- Open the ESP-IDF terminal and enter `idf.py menuconfig`. Configure as shown below, selecting the corresponding `OV5640` camera.



- Enable the DVP interface.



- Save the configuration. Recompile and download.

Q: Why does compilation fail?

- If the following situation occurs when compiling a project in Arduino IDE, you should modify `Partition Scheme` -> `16M Flash (3MB APP/9.9MB FATFS)`.

```
Sketch uses 1404340 bytes (107%) of program storage space. Maximum is 1310720 bytes.
Global variables use 96644 bytes (29%) of dynamic memory, leaving 231036 bytes for local variables. Maximum is 327680 bytes.
Sketch too big; see https://support.arduino.cc/hc/en-us/articles/360013825179 for tips on reducing it.
text section exceeds available space in board

Compilation error: text section exceeds available space in board
```

- When an ESP-IDF project fails to compile, delete the `build`, `managed_components`, and `dependencies.lock` folders, then compile again. If the error persists, check the `ESP-IDF` version and whether the `CMakeList.txt` in the project directory is recognized correctly.

Q: The module keeps resetting?

- This situation is caused by a program crash that makes USB malfunction. Press and hold the BOOT button while powering on; the module will enter download mode. Flashing the firmware (program) will resolve the issue.

Q: The first program compilation is extremely slow. How to handle it?

- It's normal for the first compilation to be slow, just wait patiently.

Q: After successful flashing in ESP-IDF, the serial port shows 'waiting for download...'. What should I do?

- If the development board has a reset button, press it; if not, power cycle the board.

Q: How can I measure the battery voltage on this product?

- This product includes an onboard `AXP2101` that can measure various battery parameters, such as chip temperature, charging status, whether power is connected, battery voltage, charging voltage, battery capacity, etc.

Q: Why does the battery percentage reading fluctuate significantly on this product?

- The `AXP2101` estimates battery capacity based on battery voltage, but battery capacity is not linear. Therefore, percentage readings may fluctuate considerably, especially when plugging/unplugging the charger, under load changes, or as the battery ages.

The measured percentage is only a reference; voltage should be used as the actual measure of battery level.

Q: Can you provide the 3D CAD file of the case?

- No.

Q: Are both buttons on this product programmable?

- Yes. The two buttons on this product are BOOT and PWR, with the following functions:
  - `BOOT`: If pressed while powering on, the development board enters download mode (commonly used when the program freezes or the USB GPIO is occupied). Under normal operation, the button state can be read via `GPIO0` (low level indicates pressed). It can detect single-click, double-click, multi-click, and long-press actions.
  - `PWR`: When powered on, long-press for 6 seconds to power off. When powered off (the device can still charge the battery), single-click to power on. Under normal operation, the button state can be read via the expansion IO `EXIO6` (high level indicates pressed). It can detect single-click, double-click, multi-click, and long-press actions (long-press must not exceed 6 seconds, otherwise the device will power off).

Q: Does this product support Bluetooth?

- The core chip of this product is the ESP32-S3R8, which supports 2.4GHz Wi-Fi, low-power Bluetooth 5, and other wireless communication functions. It includes an

onboard chip antenna with optimized gain to support Bluetooth and Wi-Fi. For details, please refer to ESP32-S3 datasheet.

Q: Does this product support Chinese character display?

- This product uses the ESP32-S3R8 as the core and connects to the screen via QSPI. It fully supports Chinese character display, provided the following conditions are met:
  i. Character encoding support: strings in UTF-8 encoding can be rendered to the screen.
  ii. Font support.
  iii. Display library support: both LVGL and Arduino_GFX support Chinese display.

Q: Which chips are used for the screen on this product?

- AXS15231B, a display driver all-in-one chip.

Q: Is this product waterproof?

- No, it is not waterproof. Keep it dry during use.

Q: How can I check which COM port I am using?

Windows System:

1. Through Device Manager: Press `Windows` + `R` to open the "Run" dialog box. Type `devmgmt.msc` and press Enter to open Device Manager. Expand the "Ports (COM &

LPT)" section. All COM ports and their current status will be listed here.

2. Using Command Prompt: Open the Command Prompt (CMD), type the `mode` command, which will display status information for all COM ports.
3. Check the hardware connection: If an external device is already connected to a COM port, the device typically occupies a port number. You can determine which port is being used by checking the connected hardware.

Linux System:

1. Check using the `dmesg` command: Open the terminal.
2. Check using the `ls` command: Type `ls /dev/ttyS*` or `ls /dev/ttyUSB*` to list all serial devices.
3. Use the `setserial` command: Type `setserial -g /dev/ttyS*` to view configuration information for all serial devices.

Q: Why is there no output on the monitor after successful programming and running, even though the code is fine?

- The Type-C programming/debugging port on the ESP32-S3-Touch-LCD-3.5B is directly connected to the ESP32-S3 USB. In the Arduino IDE development environment, `printf` output is supported. To also support `Serial` output, you need to enable the `USB CDC On Boot` function or declare `HWCDC`. Refer to the demonstration in the example code. In `ESP-IDF`, `ESP_LOGD` and `ESP_ERROR_CHECK` are typically used for output.

Q: Can you help me review or modify my code? Can you help me modify the code?

This product is positioned as a development board, not a finished product. This product is positioned as a development board, not a finished product. The product ecosystem is based on the ESP32 core, which is very mature and the development environment is very

friendly. We do not assist in modifying code. Please let the makers and geeks use their DIY skills. If you have questions, you can ask our engineers for answers.

If you like our product and wish to customize hardware, casing, software, etc. in bulk, please feel free to contact our **sales**.



Q: How do I design interfaces using `SquareLine Studio`?

- Please refer to **SquareLine Studio Tutorial**.

Give Feedback